

# Matthew Cromer

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## Education

**University of Connecticut** – *B.S. in Mechanical Engineering - 3.6/4.0 GPA*

August 2023 - May 2027

- Dean's List Spring 2026 with a 3.9 GPA
- Pursuing a double concentration in Dynamic Systems and Control (Robotics) as well as Design and Manufacturing

## Experience

**Mechanical Engineering Intern**, Marmon Utility – Amherst, NH

May 2026 – August 2026

- Collaborated with the molded products team on the design, testing, and technical reporting of powerline insulator components, conducting tensile, compressive, cantilevered, and wet/dry electrical flashover testing, as well as material examination with a Lumafield X-Ray Machine.
- Designed multiple mechanical clamping boot prototypes with 3D printing, one of which increased the withstood tension by 1,300% over the previous design while maintaining electrical integrity.
- Worked with an automation engineer to gain hands-on exposure to automated material handling equipment, packaging technologies, injection molding, 6-axis robotic arms, and high-speed manufacturing systems

**Landscape Laborer**, O'Neill Landscaping – Elliot, ME

May 2024 – August 2025

- Promoted from entry-level laborer to independently operating a company vehicle after my first summer, reflecting demonstrated reliability, judgment, and technical competence

**Crew Member**, Chipotle – Stratham, NH

February 2022 – December 2023

- Completed management-track training covering front & back of house, inventory, and team leadership responsibilities

## Projects

**Autonomous Robotic Car**

[Robotic Car Link](#)

- Achieved autonomous navigation and obstacle avoidance by integrating ultrasonic and IMU sensors on an Arduino-based 4WD platform.
- Programmed in Arduino/C++ to implement line-following, obstacle-avoidance, and object-following modes; used programming to prototype custom autonomous behaviors
- Diagnosed and resolved wiring and sensor-mapping issues using a custom-developed circuit schematic, reinforcing embedded systems debugging skills directly applicable to robotics/automation roles

**V6 Car Engine Design & Simulation**

[Engine Simulation Link](#)

- Modeled all individual components of a V6 engine using SolidWorks part design tools and then assembled 50+ individual components into a complete, functioning engine assembly with proper geometric constraints
- Created a motion study to animate the assembled engine, visualizing piston-crankshaft interaction and cylinder firing sequences.

**Drone Structural Analysis**

- Performed finite element analysis on a quadcopter drone frame using ANSYS Mechanical and SpaceClaim
- Modeled realistic loading conditions, including thrust, gravity, and aerodynamic drag, to evaluate structural performance under real-world flight and landing scenarios, as well as performing a failure analysis.
- The result of this analysis was a new design with decreased weight, leading to lower costs.

## Skills

**CAD:** SolidWorks CAD/CAM, SpaceClaim, Discovery, Technical Drawings (GD&T), MoldFlow

**Simulation & Analysis:** ANSYS Fluent & Mechanical, Simulink, LabVIEW, SolidWorks Analysis

**Programming & Systems:** MATLAB, Python, C++, HTML, Arduino, Raspberry Pi, MS Excel, VS Code

**Fabrication & Testing:** Additive manufacturing/3D Printing, Mechanical (tensile, bending, torque), Lean Principles, Failure Mode and Effects Analysis (FMEA), Injection Molding